TinyAlign: Cross-Attention Bridges for Efficient Distillation

C. Scientist, D. Student — Demo Lab (Sample Paper B)

Abstract

Compact student networks often lose accuracy when distilled from large teachers. We study TinyAlign, a distillation recipe that aligns intermediate representations through a lightweight cross-attention bridge instead of direct feature matching.

On GLUE-Mini, a reduced 4-task subset of GLUE, TinyAlign recovers 87.4% of the teacher accuracy while running 2.6 times faster than the original encoder.

Introduction

Knowledge distillation typically minimises the distance between teacher and student hidden states, yet these representations live in different geometric spaces when the student is much smaller. Prior work compensates with deep projection layers that add latency of their own.

We propose a cross-attention bridge that lets the student query the teacher's representations at run time only during training, so inference cost is unchanged while the alignment signal is richer.

Methodology

Experiments use GLUE-Mini, a 4-task subset (MNLI-m, SST-2, QQP, STS-B) with 15% of the original training data. Students are 4-layer and 6-layer encoders distilled from a 12-layer teacher, and all runs repeat over three random seeds with mean and standard deviation reported.

The bridge uses 8 attention heads and is discarded after the 20-epoch distillation schedule. Latency is measured on the same Jetson Orin class of hardware used in related pruning studies.

Results

TinyAlign recovers 87.4% of teacher accuracy on GLUE-Mini on average across the four tasks, compared with 84.9% for direct feature matching under an identical student size and budget.

The bridge adds 12 minutes to distillation but nothing at inference; student latency measured 7.9 ms per sentence, a 2.6 times speed-up over the teacher at batch size one.

Limitations

Because GLUE-Mini keeps only 15% of the training data, absolute numbers are not directly comparable to full-benchmark results, and conclusions about very small students (fewer than 4 layers) remain outside the scope of this study.